

4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Input Parameters
```

```
fp = 1000;      % Passband cutoff frequency (Hz)
```

```
Fs = 4000;      % Sampling frequency (Hz)
```

```
N = 50;         % Filter order
```

```
% Normalized cutoff frequency
```

```
Wn = fp/(Fs/2);
```

```
%% Design FIR Low-Pass Filter using Hamming Window
```

```
b = fir1(N, Wn, 'low', hamming(N+1));
```

```
%% Display Filter Coefficients
```

```
disp('FIR Low-Pass Filter Coefficients (Hamming Window):');
```

```
disp(b');
```

```
%% Frequency Response
```

```
figure;
```

```
freqz(b,1,1024,Fs);
```

```
title('Frequency Response of FIR Low-Pass Filter (Hamming Window)');
```

```
grid on;
```

```
%% Magnitude Response
```

```
[H,f] = freqz(b,1,1024,Fs);
```

```
figure;
```

```
plot(f,20*log10(abs(H)),'LineWidth',2);
```

```
grid on;
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude (dB)');
```

```
title('Magnitude Response');
```

```
axis([0 Fs/2 -100 5]);
```

```
%% Phase Response
```

```
figure;
```

```
plot(f,unwrap(angle(H)),'LineWidth',2);
```

```
grid on;
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Phase (radians)');
```

```
title('Phase Response');
```

```
%% Impulse Response
```

```
figure;
```

```
impz(b,1);
```

```
grid on;
```

```
title('Impulse Response');
```

```
%% Pole-Zero Plot
```

```
figure;
```

```
zplane(b,1);
```

```
grid on;
```

```
title('Pole-Zero Plot');
```

```
%% Filter Information
```

```
fprintf('Filter Order = %d\n',N);
```

```
fprintf('Cutoff Frequency = %d Hz\n',fp);
```

```
fprintf('Sampling Frequency = %d Hz\n',Fs);
```









